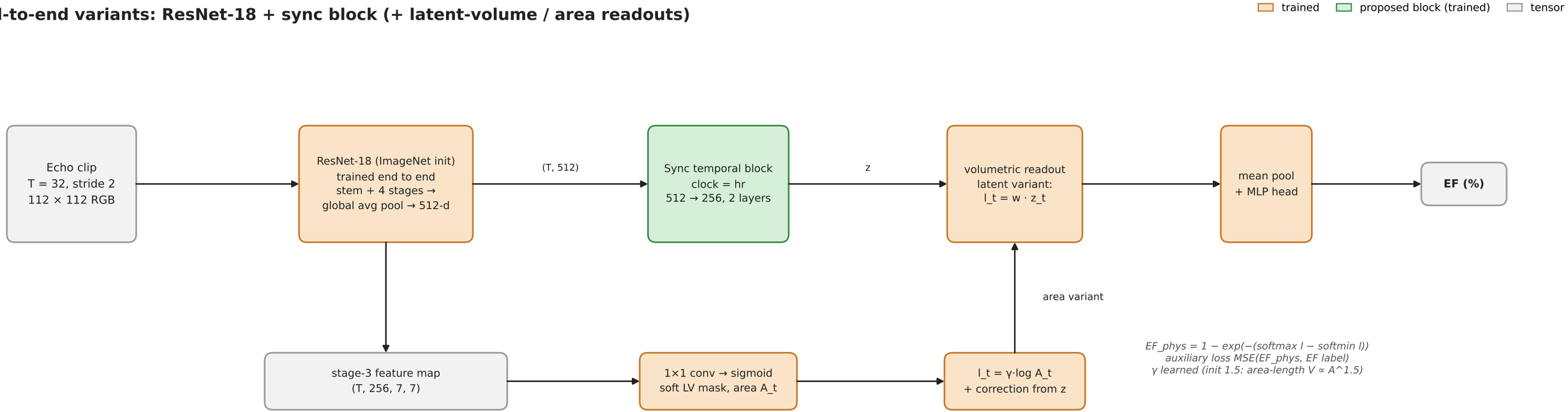
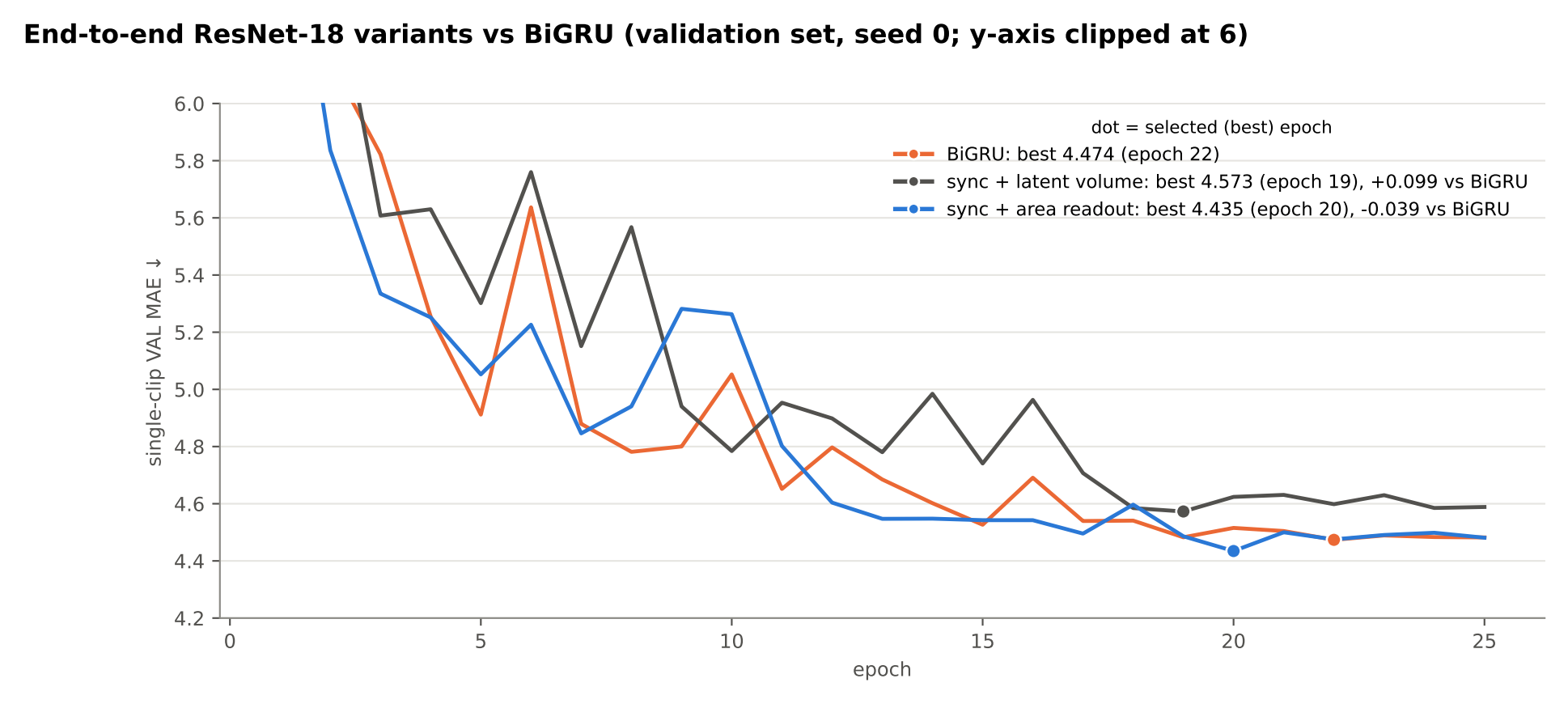
End-to-end variants: ResNet-18 + sync block (+ latent-volume / area readouts)



First track (before the frozen-PanEcho track): same clip protocol, pooling and head as the proposed model; the area variant adds the mask branch.


End-to-end ResNet-18 variants vs BiGRU (validation set, seed 0; y-axis clipped at 6)
dot = selected (best) epoch
BiGRU: best 4.474 (epoch 22)
sync + latent volume: best 4.573 (epoch 19), +0.099 vs BiGRU
sync + area readout: best 4.435 (epoch 20), -0.039 vs BiGRU
single-clip VAL MAE ↓
epoch
6.0
5.8
5.6
5.4
5.2
5.0
4.8
4.6
4.4
4.2
0
5
10
15
20
25